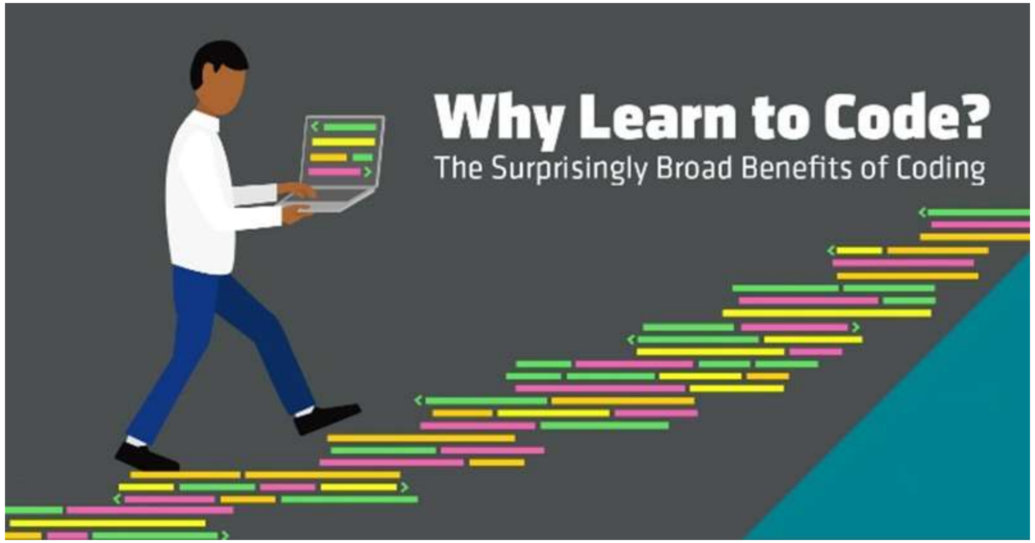
Introduction to Data Science for Geoscientists

Amr. Moslim

#GeoML



Today's Agenda

- What is Data science?
- What is Machine learning?
- Brief History on Data Science
- Data Analysis vs Data Analytics
- Data Science VS Machine learning
- Why to learn to code?
- Data Science knowledge domains
- Data Science life cycle
- Data Science Skills
- Why should I learn Python?
- How to Code?

What’s Big Data?

64,140
Instagram photos

336,480
Skype Calls

5,365,260
Youtube Videos

5,500,560
Google Searches

181,331,340
Emails Sent

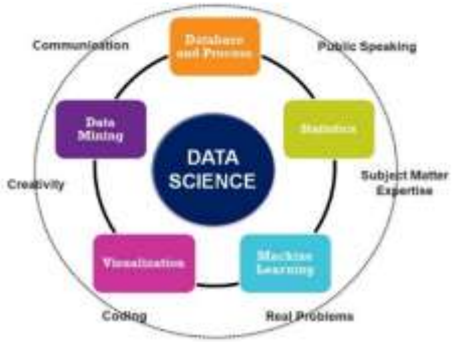
Every Single Second

2.5 Quintillion bytes of data generated Every Single Day

What's Big Data?

Big data is a term that describes large, hard-to-manage volumes of data – both structured and unstructured – that overwhelm businesses on a day-to-day basis. Big data can be analyzed for insights that improve decisions and give confidence for making strategic business decisions.

https://www.sas.com/en_us/insights/big-data/what-is-big-data.html



But it's not just the type or amount of data that's important, its what organizations do with the data that matters.

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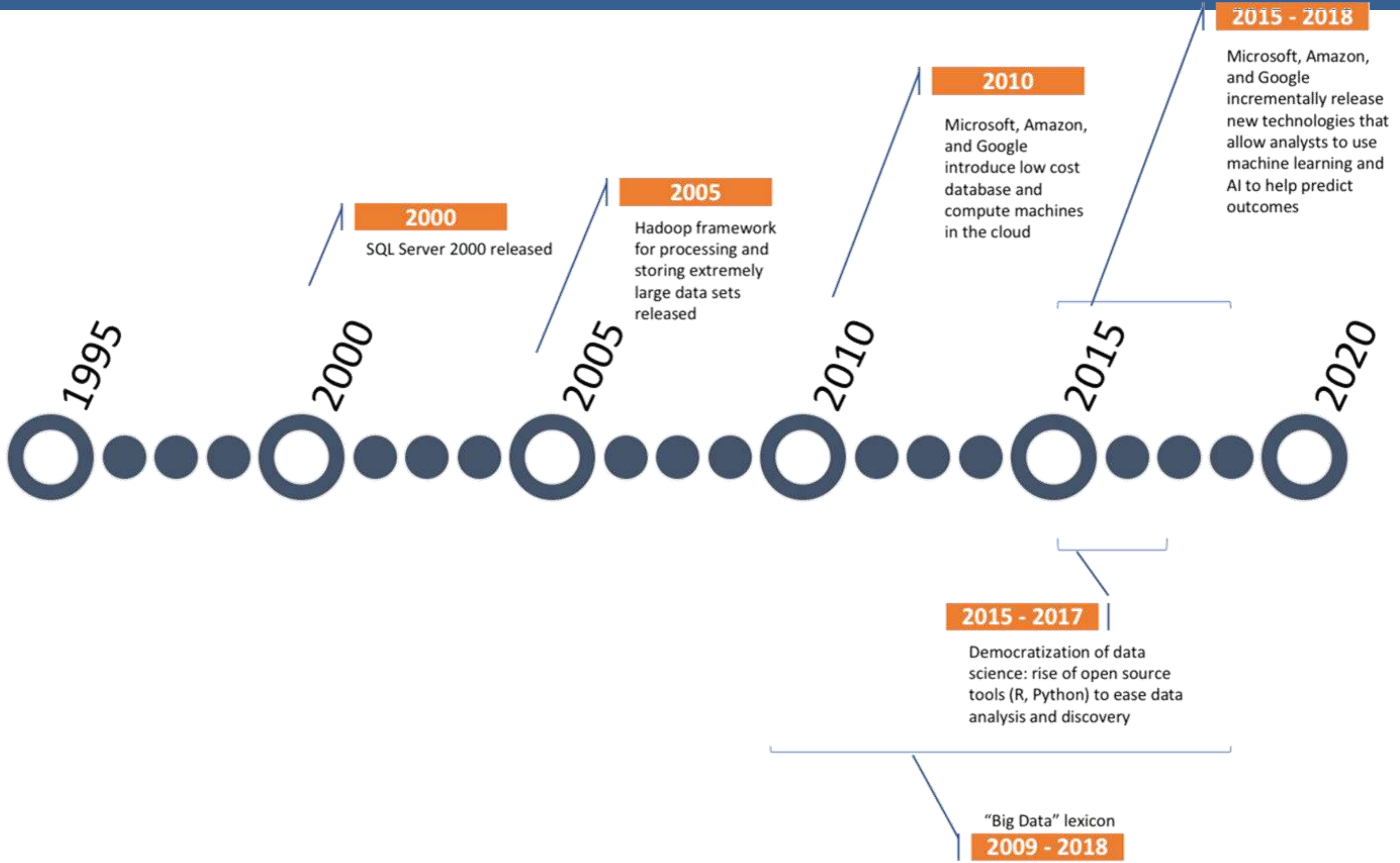
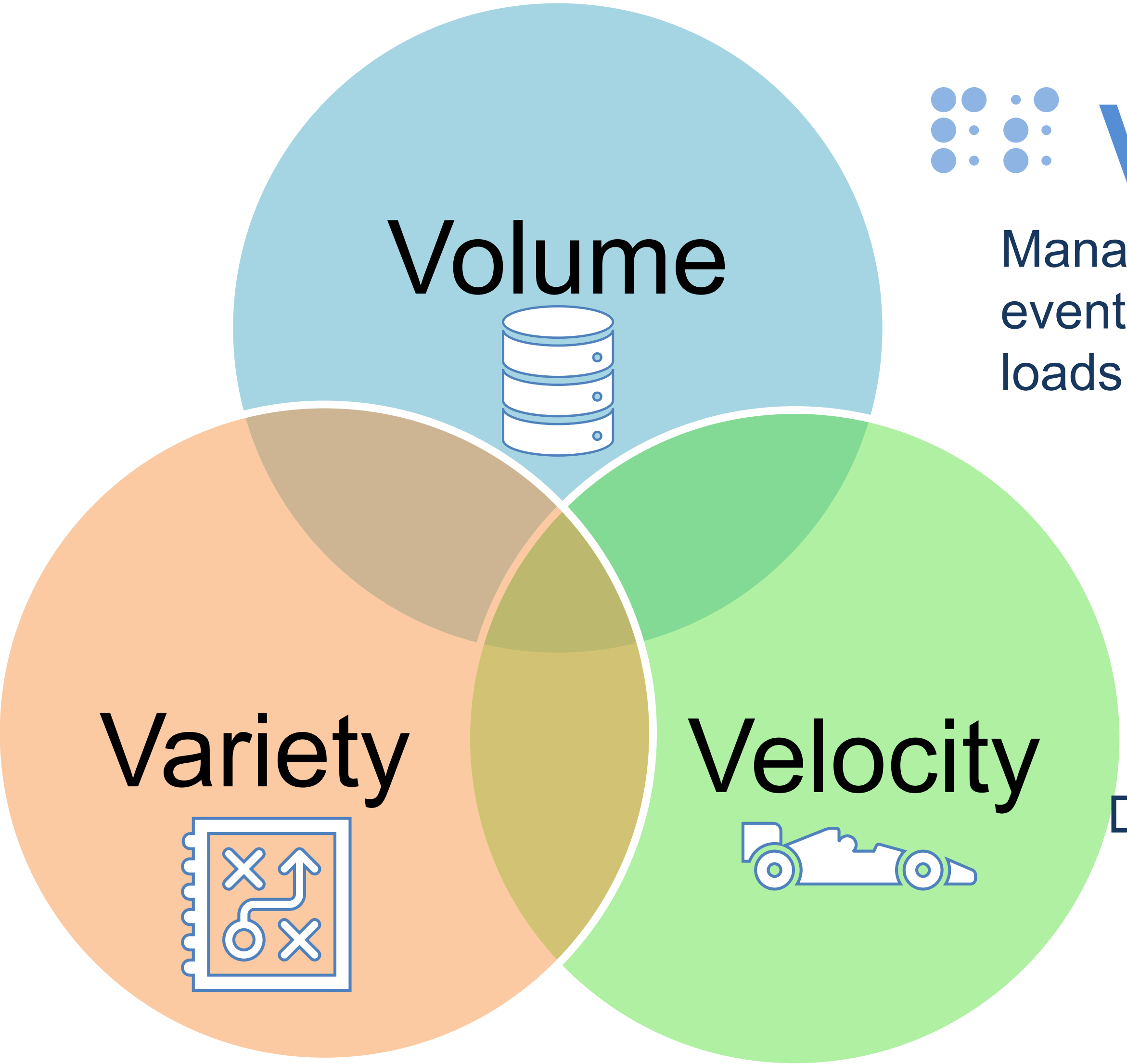
Evolution of Big Data



Big Data Challenges

Veracity

the quality of data.
Businesses need to connect and correlate relationships, hierarchies and multiple data linkages. Otherwise, their data can quickly spiral out of control.



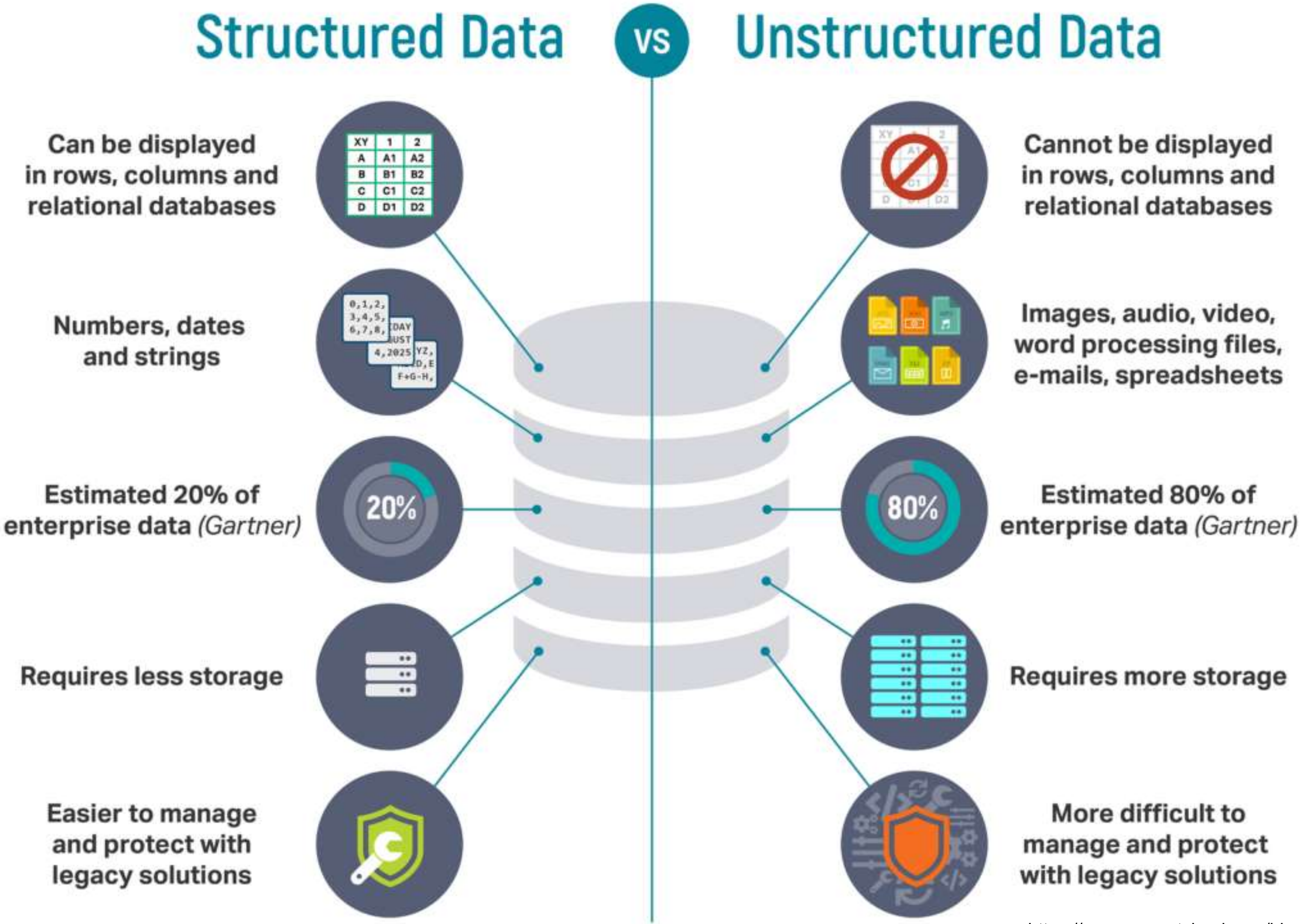
Variability

Manage daily, seasonal and event-triggered peak data loads

Velocity

Data Creation & Acquisition Growth

Big Data Types



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https://www.promptcloud.com/blog/big-data-evolution-technology-

Why Big Data is Important

- Enable smart decision making.
- Improve and fast track the operational efficiencies
- Optimize product development
- Drive new revenue and growth opportunities
- Streamline resource management
- Automate tasks that takes more time and resources so the individual can focus on the intelligence part

https://www.promptcloud.com/blog/big-data-evolution-technology-

Big Data – Popular use cases

- Internet of Things
- Information Security
- Gathering rich insights about business
- Data warehouse optimization
- Improving healing and public health

https://www.promptcloud.com/blog/big-data-evolution-technology-modern/

Big Data – Oil & Gas industry Use Cases

- Gathering real time information while drilling
- Production monitoring sensors
- Seismic data acquisition products
- Geoscience data (well logs, cores, maps, remote sensing images)
- Well test data and reservoir monitoring data

Data Storage and Management tools

Data Cleaning and Processing tools

"Deep learning craves big data because big data is necessary to recognize hidden patterns and to find answers without overfitting the data. With deep learning, the more good quality data you have, the better the results"

ATLAS.ti

python™

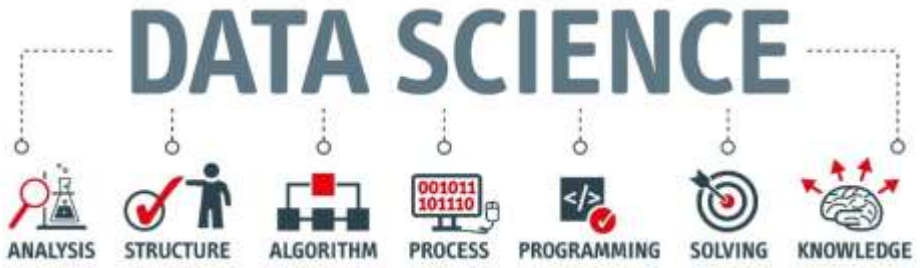
R

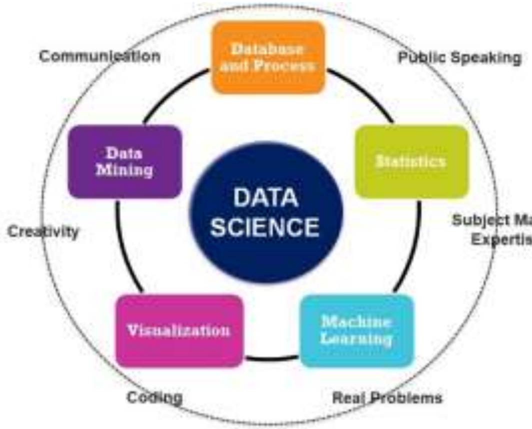
OpenRefine

Time For Break

What's Data Science ?

Data science is the field of study that uses modern tools and techniques to process, clean, analyze, model and visualize large data sets to get insights that are reliable to help organizations to understand certain criteria or condition and make business decisions





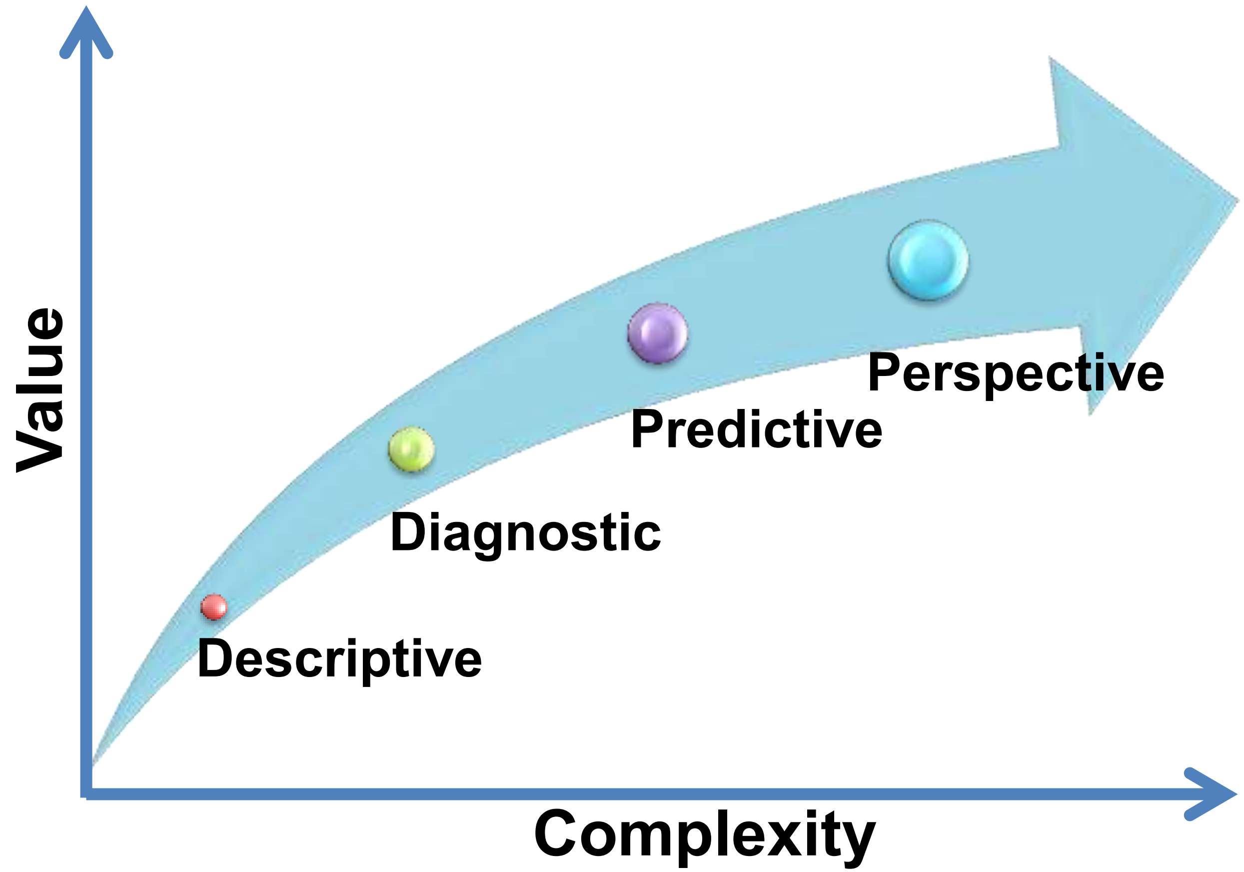
Data Science History

Year	Milestone	Details
1962	The future of Data analysis	**John W. Tueky** released a paper about the future and importance of Data Analysis
1974	Computer methods	**Peter Naur** used term Data Science for the first time
1977	Exploratory Data Analysis	**John W. Tueky** Introduced the term of EDA
1989	Data bases Workshop	**George Shapiro** Introduced a workshop about data bases
1999	The need to have new devices for enormous data	**Jacob Zahavi** Introduced the need to power devices for the enormous data "Big Data"
2001	International Council for science and technology	Committee on Data for science and Technology
2008	Title "Data Scientist" released	A trendy expression for who works in data analysis and analytics

Data Analysis

- Data Preparations
- Data Cleaning / Editing
- Statistics: Frequencies, Means, Standard Deviation, Correlation , Probabilities , Variances, Scaling, Standardization Methods, Outlier removal
- Visualizations : Univariate, Bivariate, Multivariate
- Interpretations of Trends and Patterns Insights

Data Analytics



Descriptive "What's happening" :

- Data understanding & Exploration
- Data visualization

Diagnostic "Why it is happening":

- Dive into the root cause
- Isolate all factors and eliminate noise

Predictive "What's going to happen":

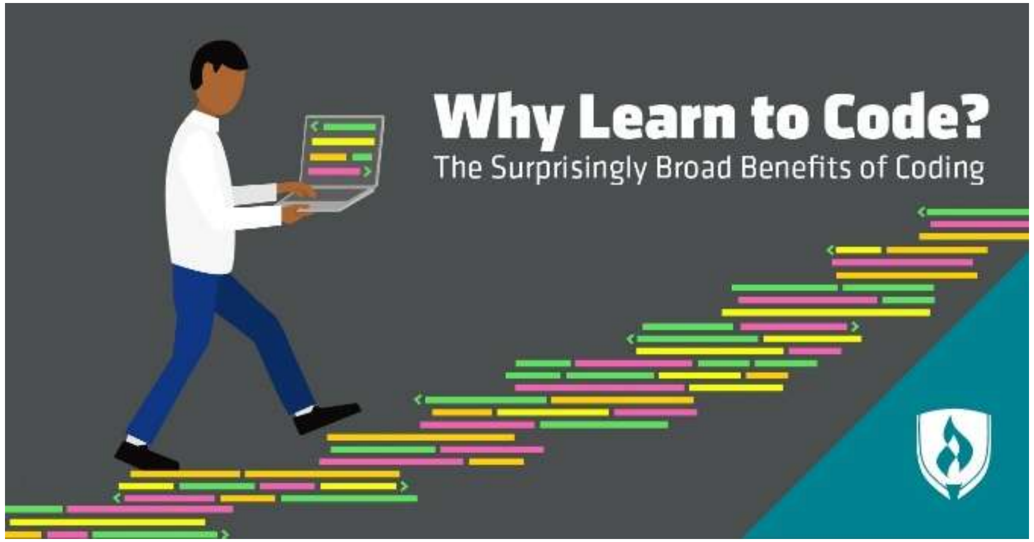
- Historical patterns are used to predict specific future outcomes using algorithms
- Decisions are automated and updated using algorithms and technology

Perspective " What should I do":

- Recommended actions and strategies based on testing strategy outcomes
- Applying advanced analytical techniques to make specific recommendations

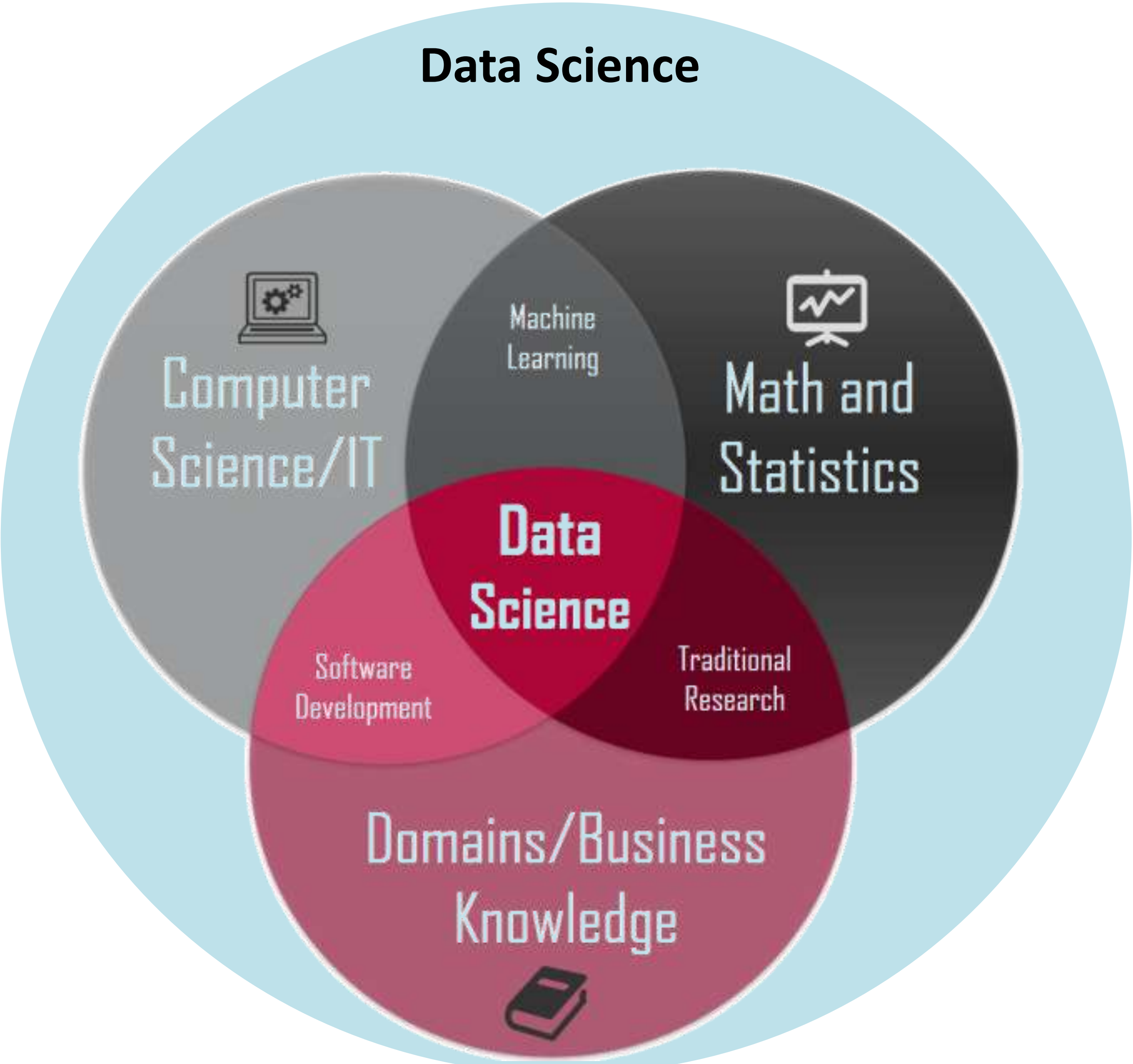
Why Learn to Code?

1. Coding is another language.
2. Coding fosters creativity.
3. Coding helps learn Math skills and makes sense of it.
4. Coding improves writing academic performance.
5. Coding can lead to software development jobs
6. It open up other job opportunities
7. Coding can make your job application stand out
8. Coding literacy can help you understand other aspects of tech
9. It could lead to freelance work
10. Coding can allow you to pursue passion projects
11. Coding can boost problem solving and logic skills
12. Coding improves interpersonal skills
13. Being a skilled coder can build confidence
14. Freedom to Make My Own projects
15. People Come to ME Asking if I Can Work for THEM
16. You can do work remotely any where.
17. I Am Part of a Top Secret Club (a.k.a., the Tech Community)
18. I Have a Sense of Self-Reliance and Empowerment



It’s a great Empowering tools and Skills

Data Science knowledge domains



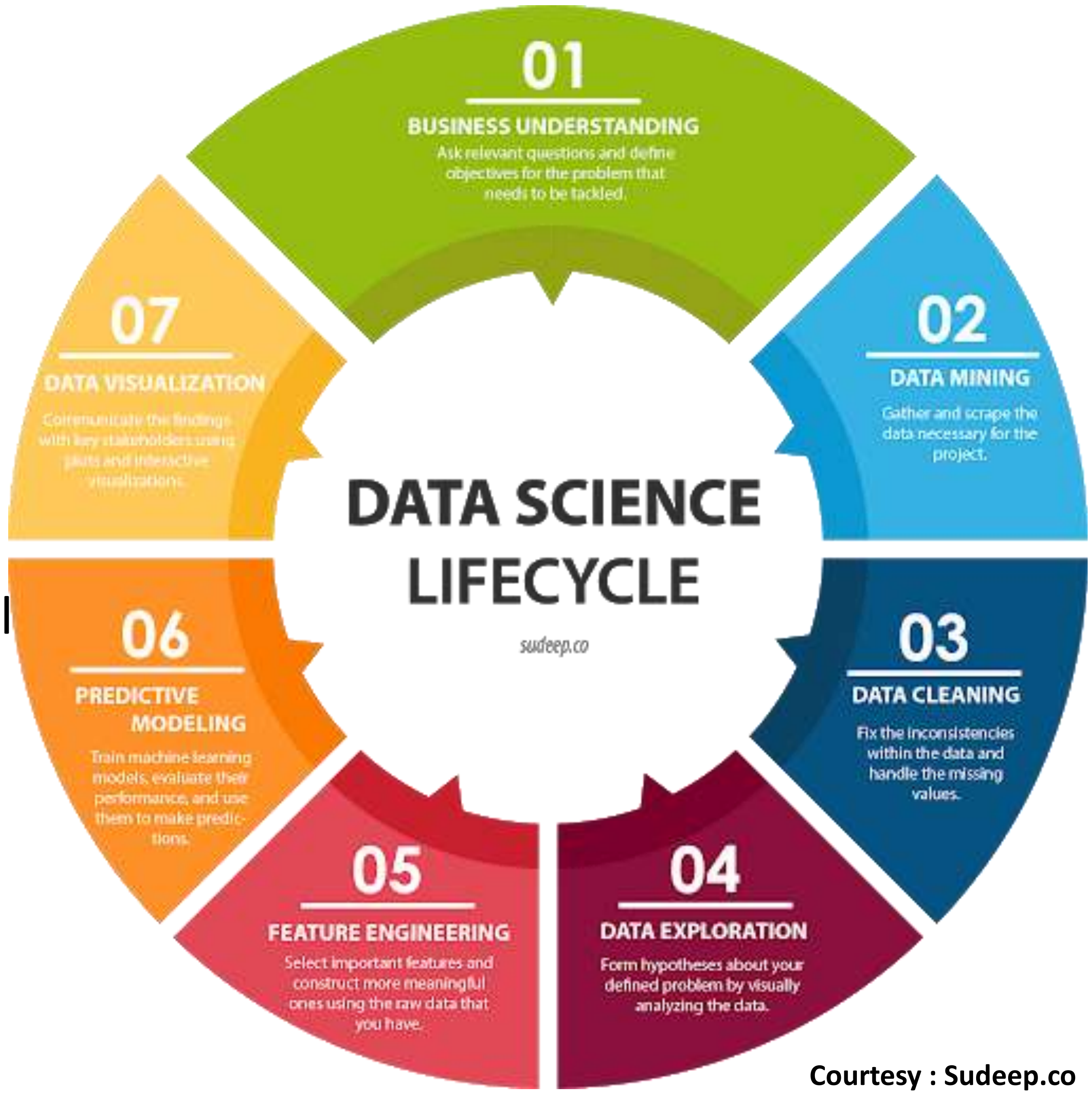
Data Science Life Cycle

80% of the Data scientist time is dedicated to

- Data collection
- Data cleaning
- Data exploration
- Feature engineering

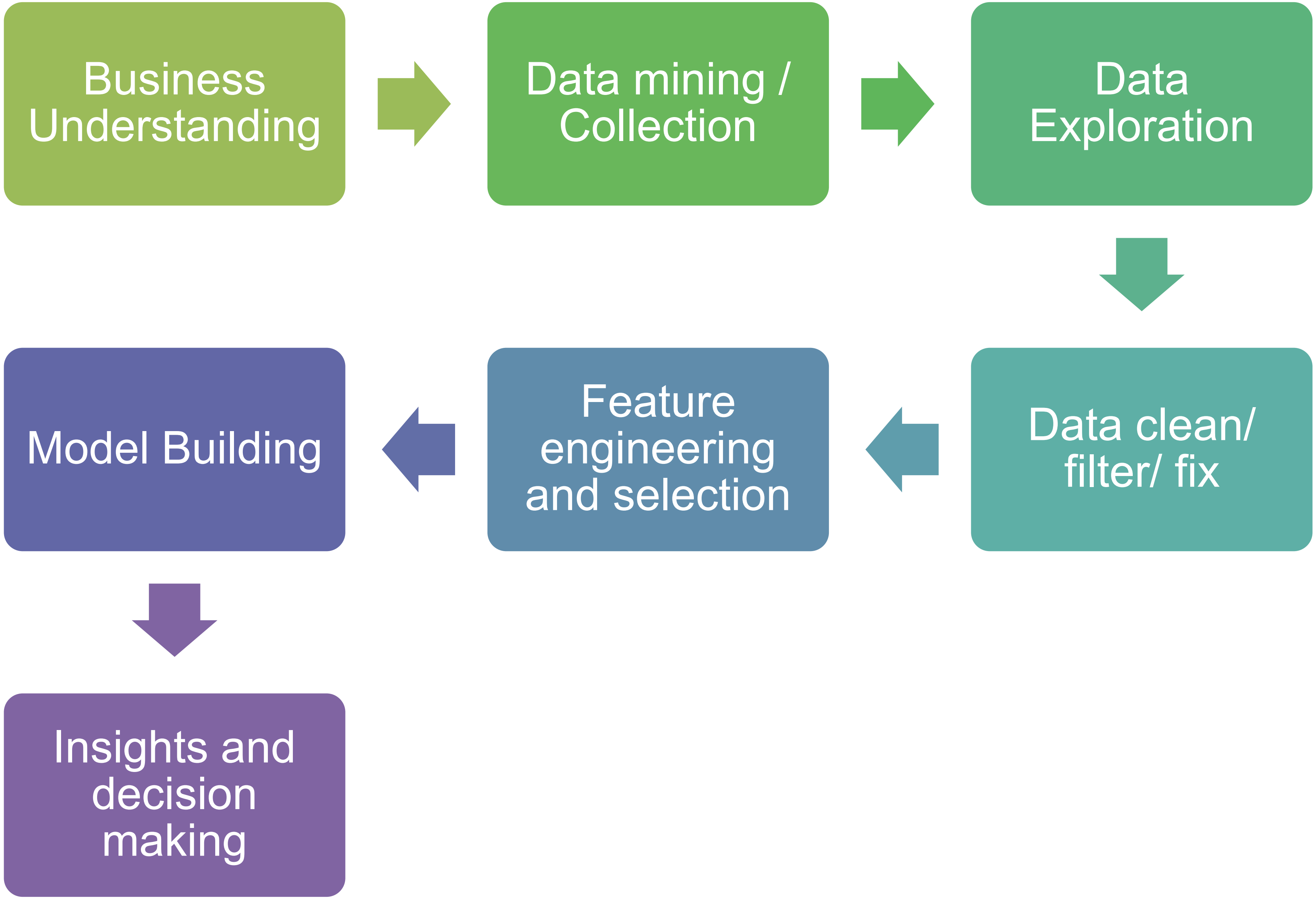
20 % of the data scientist time is for model selection and building

- ✓ Model Building
- ✓ Model Evaluation



Courtesy : Sudeep.co

Data Science Workflow



Data Science Vs Machine Learning

Data Science Vs Machine Learning

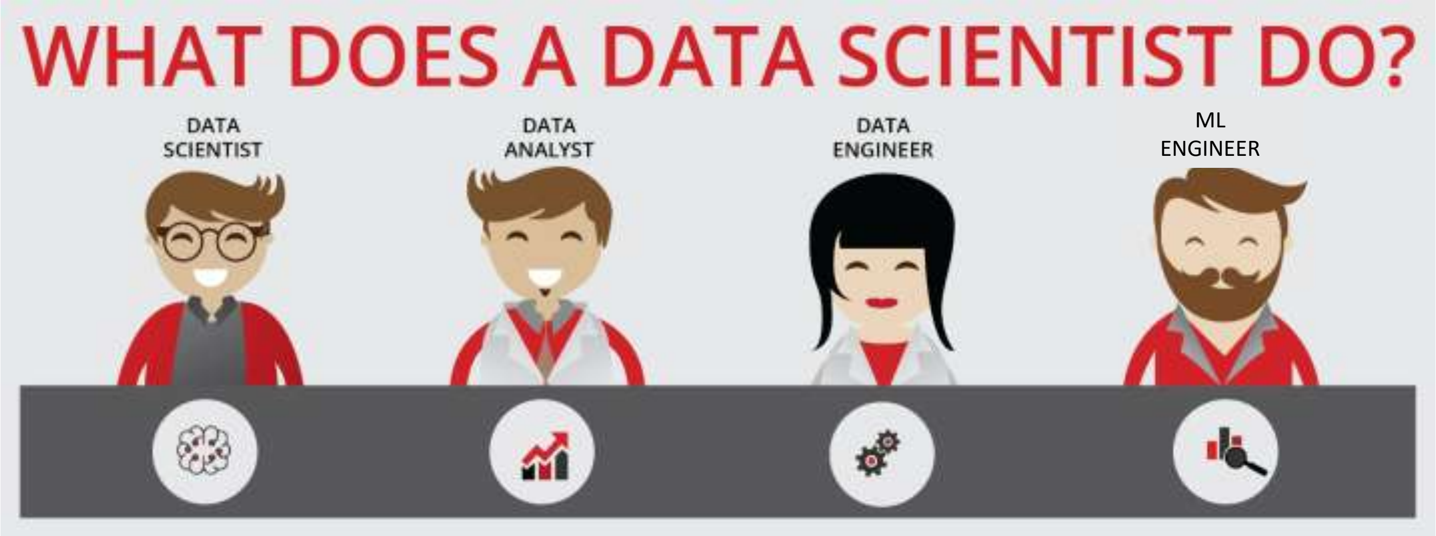
Characteristics	Data Science	Machine Learning
Objective	Focus on find unforeseen and hidden trends to understand the data pattern	Focuses on making predictions and classifications to get new data points
Tools	**Python**, **R**, SAS, Spark, Excel, MATLAB, MySQL, Tableau	**Python**, **R**, Scikit Learn, ML Studio, MS Azure
Applications in O& G	• Time series analysis • Production forecast • Oil price prediction	• S-wave log predication • Facies classification • Porosity logs prediction using seismic attributes
Skills	• Database and SQL • Mathematics and statistics • Knowledge of programming • Data mining, data wrangling • Data visualization • Machine Learning	• Programming (Python , R) • Mathematics and statistics • Machine Learning algorithms • Data Modeling • NLP

DS vs ML

Machine Learning	Data Science
Data structured - unstructured	Any type of data
No specified rules for each problem	Has specified approach and workflow for each problem
Generate generalized models for each problem type	Generate specific insights for each problem
Understanding algorithms and maths is crucial.	Domain expertise is the king
Classifies / predicts for new data points / patterns from historical data	Create insights from world complexities
Input data should be transformed specifically for the algorithm	Input data can be used directly which is to be read and analyzed

Data science Skills

Data science Jobs



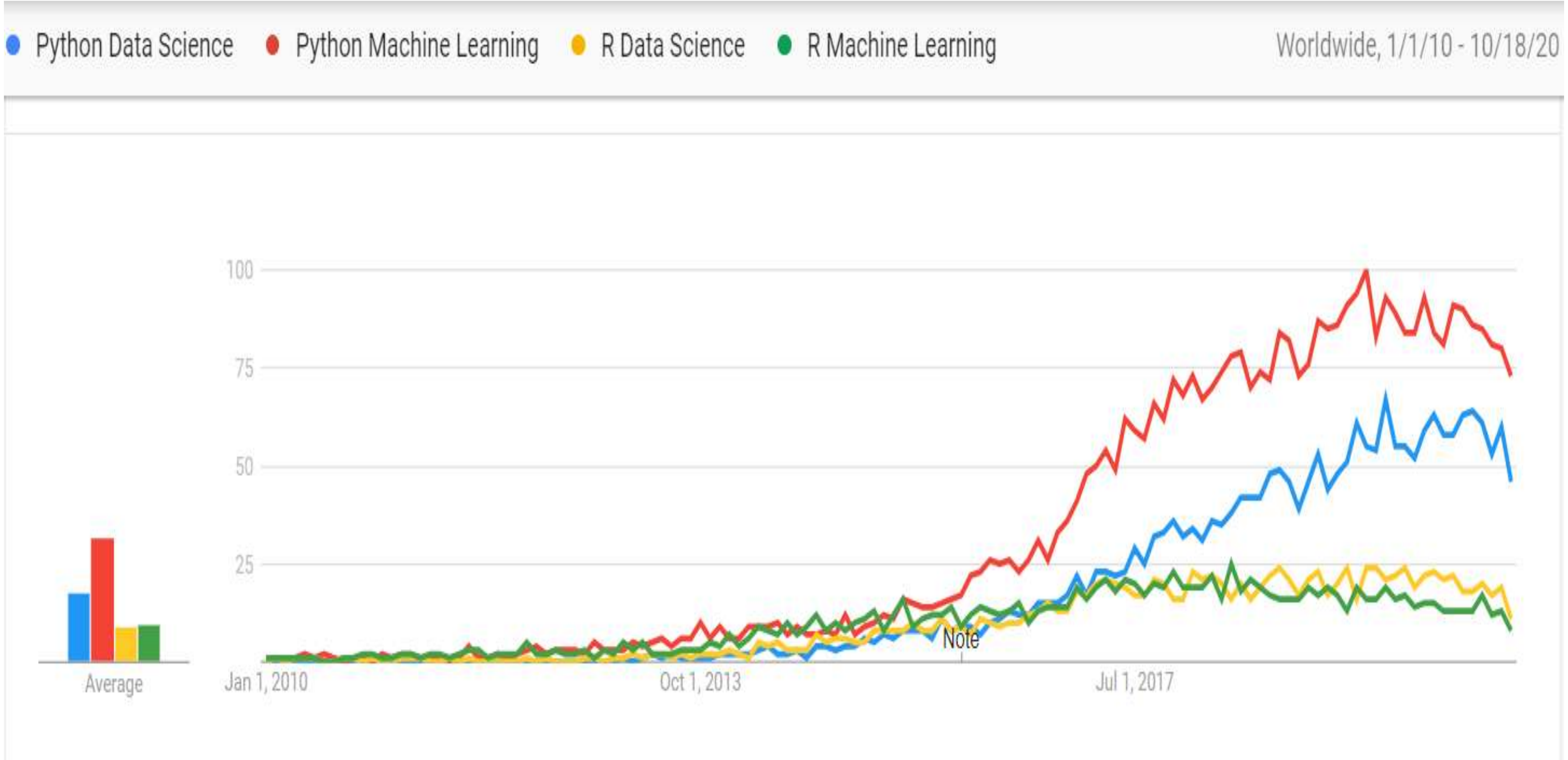
- DATA ANALYST
- DATA ENGINEER
- MACHINE LEARNING ENGINEER
- DATA SCIENCE GENERALIST

Data Scientist Skills/Experience

- **Group 1: Skills/experience related to competences**
  - Data Analytics and Machine Learning
  - Data Management/ Curation (including both general data management and scientific data management)
  - Data Science Engineering (hardware and software) skills
  - Scientific/Research Methods or Business Process Management
  - Application/subject domain related (research or business)
  - Mathematics and Statistics
- **Group 2: Big Data (Data Science) tools and platforms**
  - Big Data Analytics platforms
  - Mathematics & Statistics applications & tools
  - Databases (SQL and NoSQL)
  - Data Management and Curation platform
  - Data and applications visualization
  - *Cloud based platforms and tools*
- **Group 3: Programming and programming languages and IDE**
  - General and specialized development platforms for data analysis and statistics
- **Group 4: Soft skills or Social Intelligence**
  - Personal, inter-personal communication, team work, professional network

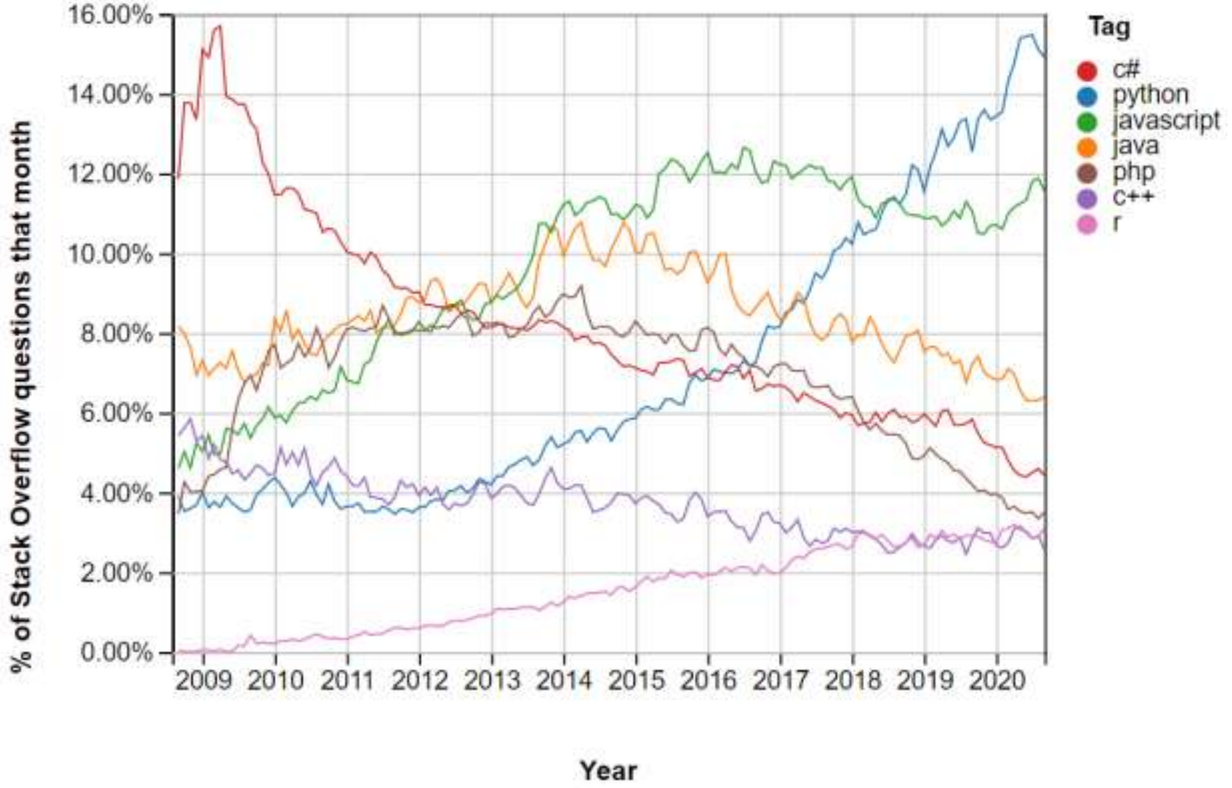
Why should I learn Python?

Why I should Learn Python ?



Why I should Learn Python ?

1. Python is the fastest growing programming language
2. Python is easy to read, write, and learn
3. Python has an incredibly supportive community
4. Open source package (free)
5. Multi purpose programming language
6. Big companies uses python in their main frame work
7. High in demand in the market of data science
8. Hundreds of applications & libraries
9. Python developers make great money
10. Great tool for reproducibility
11. Collaborative language to build complex tasks



MAJOR COMPANIES THAT USE

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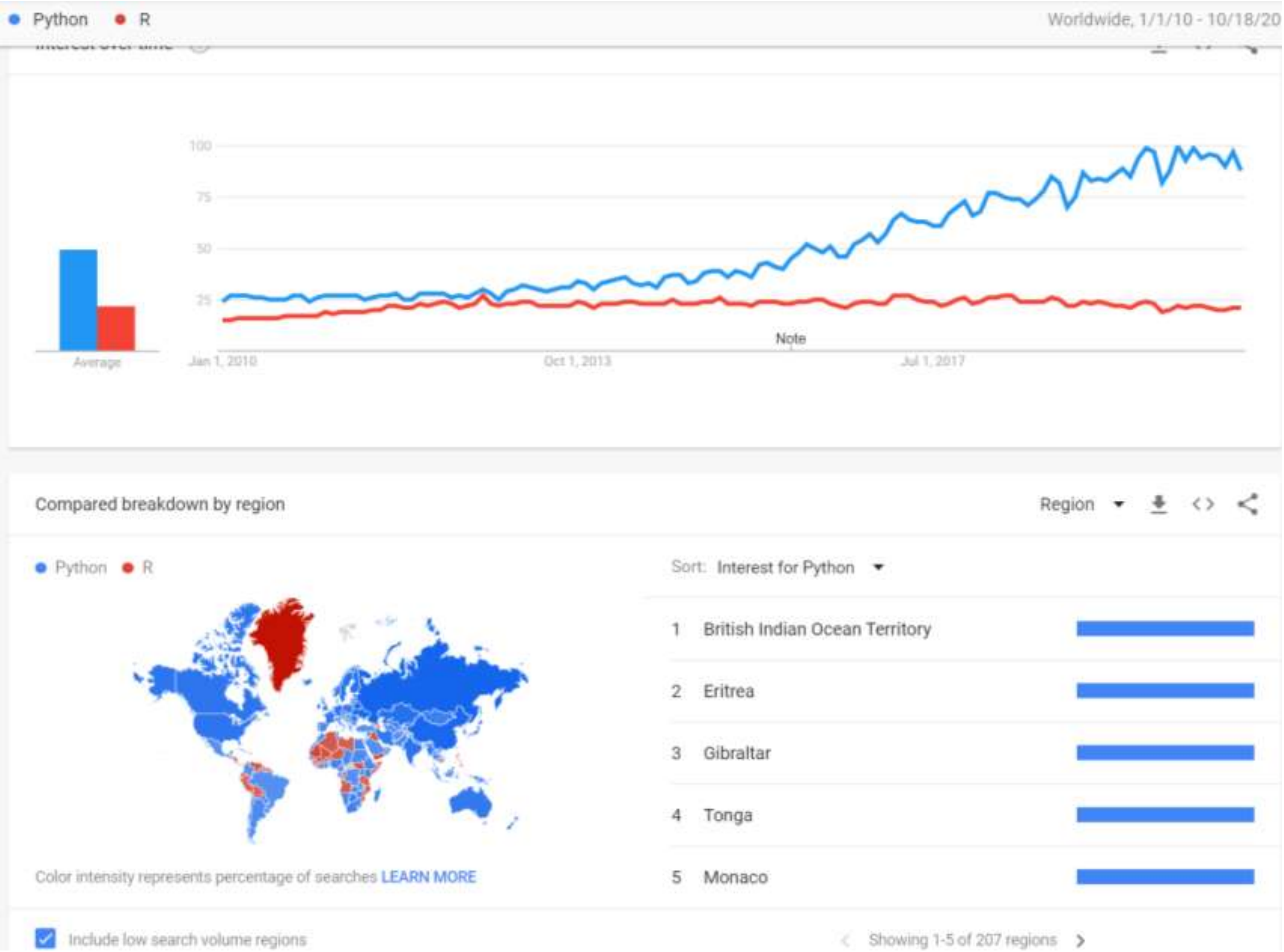
Why I should Learn Python ?



How to code?

Coding Workflow Basic Aspects

- **Assignment:**
  - ➢ Types of data structure (integer, float, String, Boolean)
- **Control flow:**
  - ➢ If statement
  - ➢ While loops
  - ➢ For loops
- **Mathematical Operators:**
  - ➢ (+, -, *, /)
  - ➢ (>, <, =, >=, <=, !=)
  - ➢ Logical operators:
    - ➢ (+=, -=, //, %, %%)
- **Functions:**

  A set of commands that works in sequence to perform a certain task that can include assignment, flow control tools and or mathematical expressions.
  - ➢ def: in Python
  - ➢ Function (x) in R
- **Error handling:**
  - ➢ Avoid having user errors
  - ➢ Handling errors
- **Reviewing:**
  - ➢ Debugging : to check that all the results as it should be even if you didn’t get any errors explicitly

Python most popular packages

- **Analysis packages**
  - **Numpy** : Numerical Manipulation and linear alegabra
  - **Pandas** : building & Manipulating DataFrames

- **Visualization packages**
  - **Matplotlib** : plots and contours
  - **Seaborn** : beautiful plots
  - **Plotly** : interactive plotting

- **Machine Learning packages**
  - **Tensorflow** : Neural NetWork and Deep learning
  - **Keras**: ML algorithms
  - **Scikit Learn**: ML algorithms and model evaluations

- **Scientific packages**
  - **Scipy** : scientific equations in python
  - **Obspy** : seismic manipulation and reading segy

- **Geoscience Package**
  - **Welly** : reading / write well logs las files
  - **Lasio** : reading / write well logs las files
  - **Segyio** : seismic Segy files reading / writing and manipulation.
  - **Petropy** : Petrophysical evaluation

Time For Break